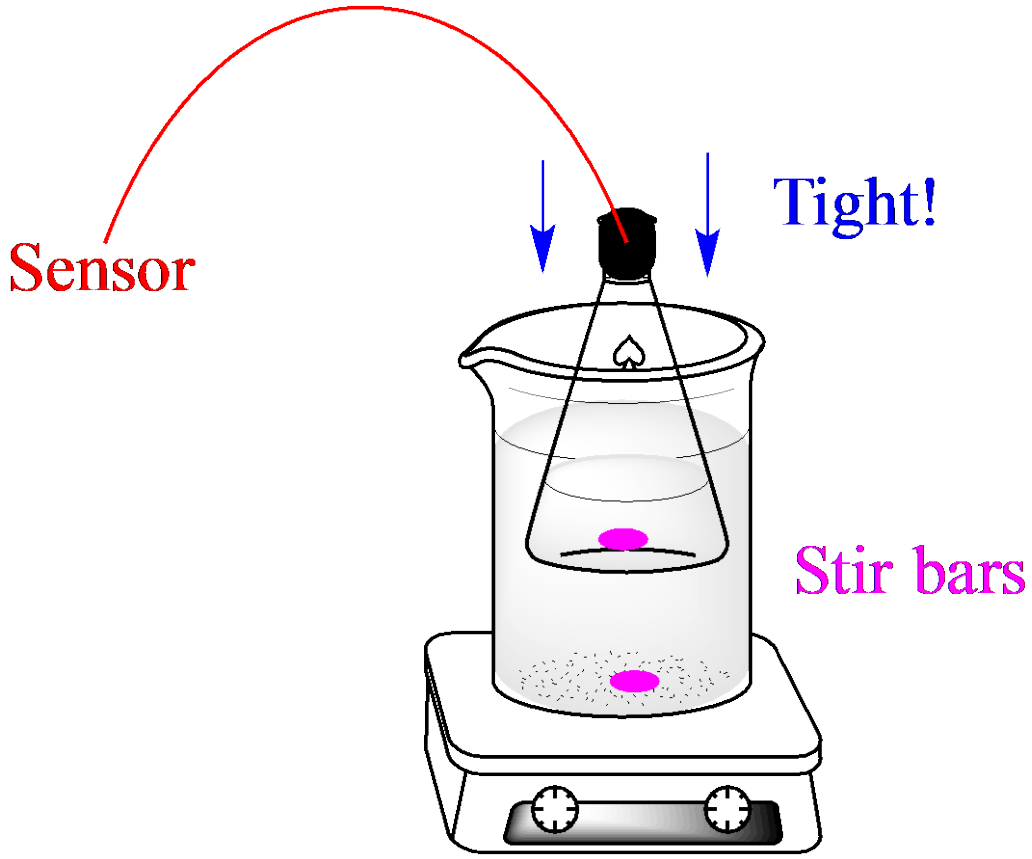
Kinetics Part II

Chem 151 Lab

Use only Mix 1 today!

	Mix 1	Mix 2	Mix 3
0.10 M KI (mL)	15.0	15.0	30.0
DI H_2O (mL)	35.0	25.0	20.0
0.88 M H_2O_2 (mL)	10.0	20.0	10.0
Total volume (mL)	60.0	60.0	60.0

$$2\,H_2O_2(aq) \xrightarrow{I^-} 2\,H_2O(l) + O_2(g)$$



Assume that:

$$Rate = k[H_2O_2]^1[I^-]^1$$

3 Runs Today:

1. Room Temp
2. 10 °C
3. 30 °C

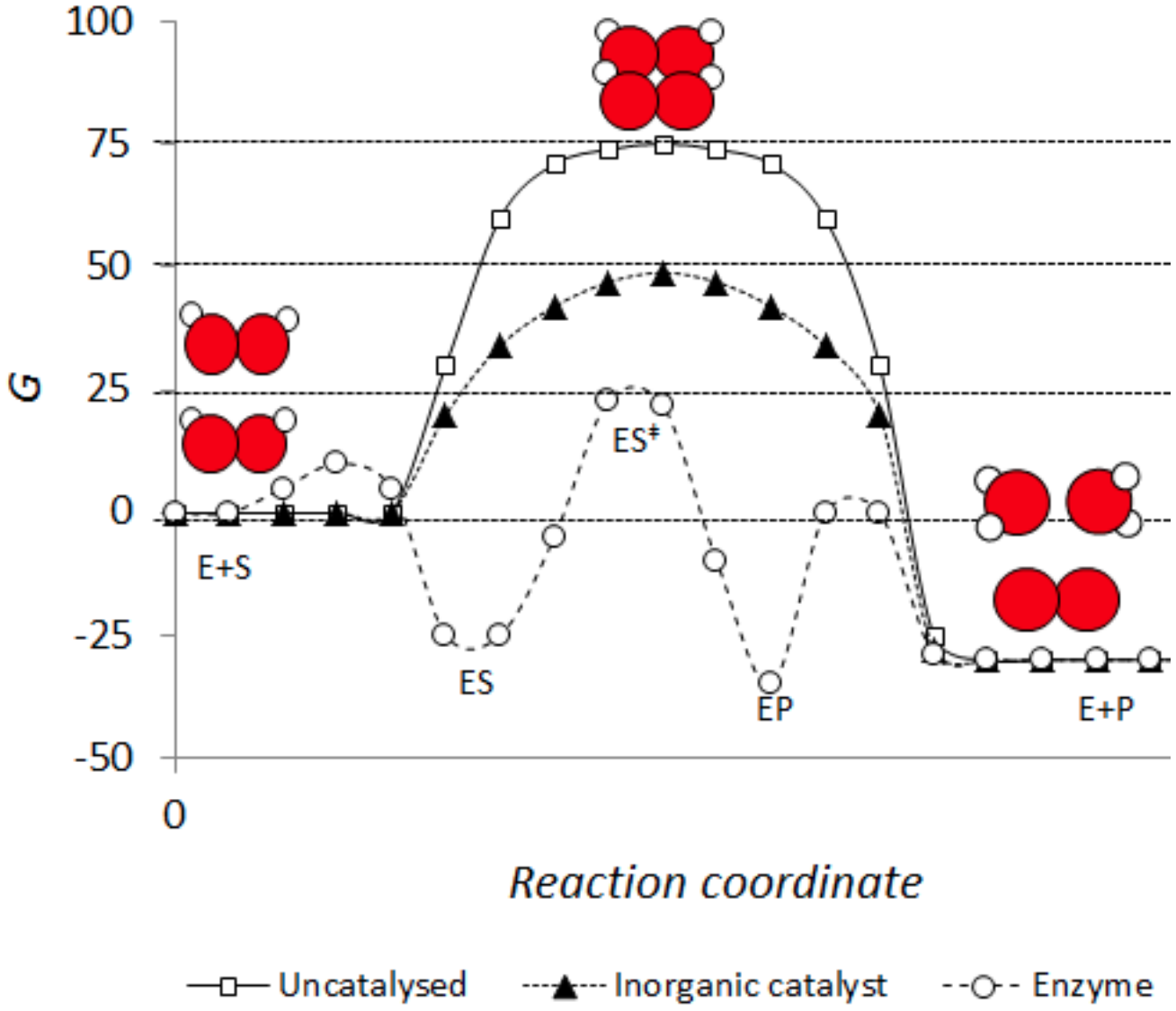
Don't get too cold!

Use hot water from taps

Also make sure you get all your solutions from the 10 °C or 30 °C baths!

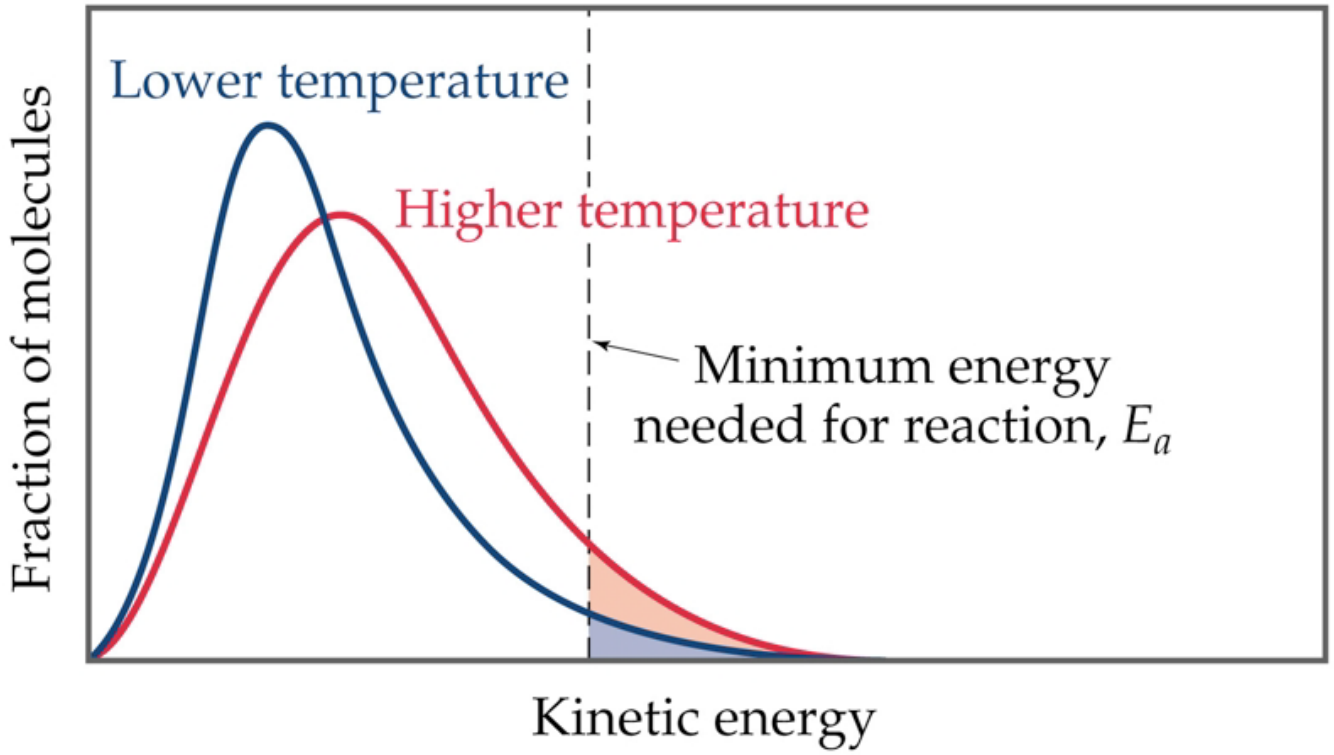
Activation Energy - E_a

Energy required to overcome the barrier to a reaction



How Temperature Impacts Rate

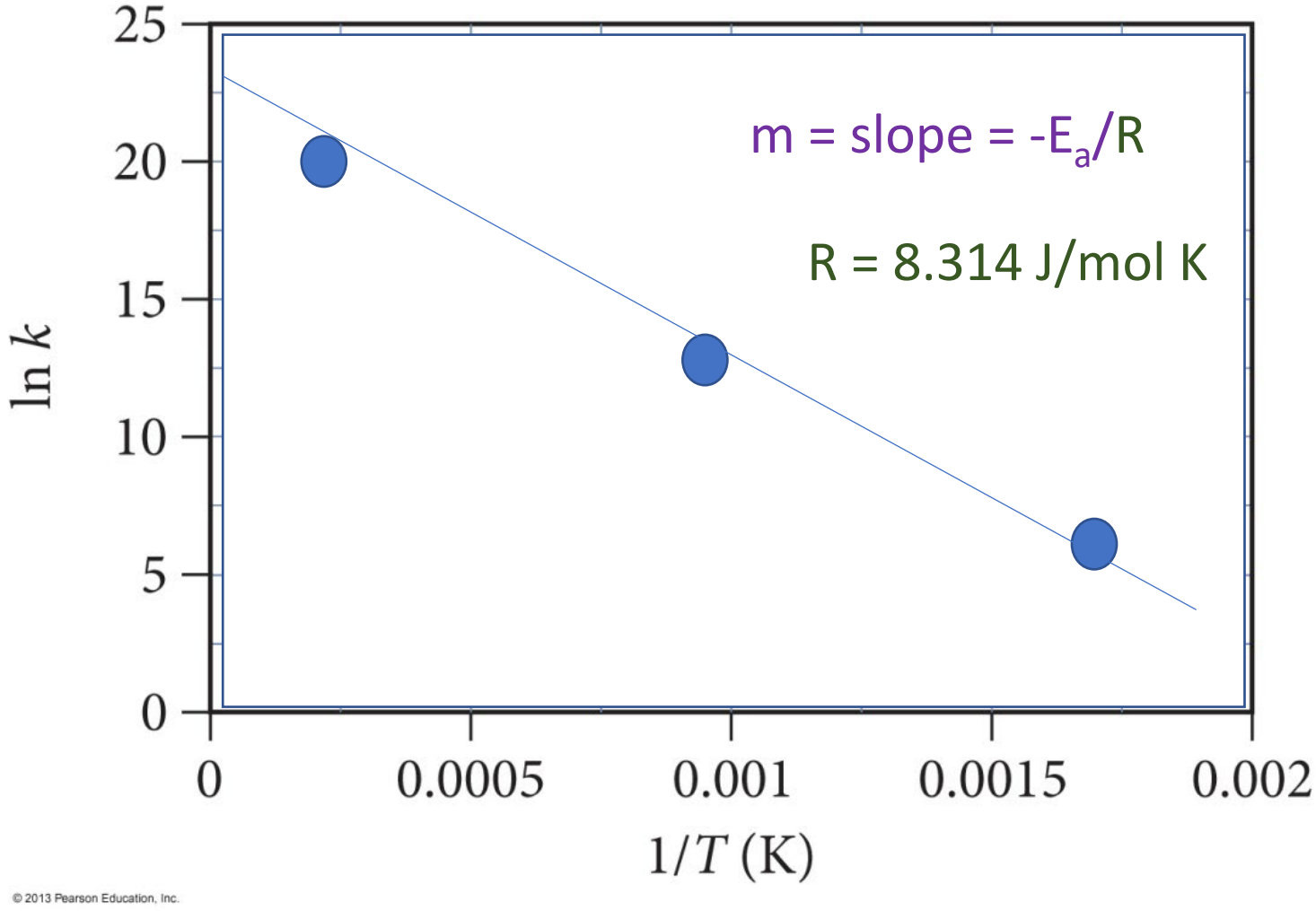
As temperature increases the number of molecules with sufficient energy to react increases



Arrhenius Equation

$$lnk = -\frac{E_a}{R}\left(\frac{1}{T}\right) + lnA$$

y = mx + b



© 2013 Pearson Education, Inc.

Reminders

- This is your last lab notebook upload! Due by 11:59pm on Sunday
- Your Lab Final Practical Exam is next week and worth 75 points!
  - What time are you coming next week??
- Please take a few minutes to fill out the course evaluations for Chem 151 Lab!! I greatly value your input.